Part A: Fundamentals

1. What is Data Analysis?

Data Analysis is the process of collecting, cleaning, transforming, and interpreting data to discover useful information, draw conclusions, and support decision-making.

It involves:

- Inspecting data
- Cleaning and handling missing values
- Applying statistical techniques
- Finding patterns and trends
- Presenting insights using visualization tools

Example:

If a company analyzes customer purchase data, it can:

- Identify best-selling products
- Understand customer behavior
- Improve marketing strategies

In simple words, **Data Analysis converts raw data into meaningful insights.**

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2. Steps to Plan a Data Science Project

A structured Data Science project typically follows these steps:

1️⃣ Problem Definition

- Understand business objective
- Define clear problem statement
- Example: "Predict customer churn"

2️⃣ Data Collection

- Gather data from databases, APIs, CSV files, etc.

3️⃣ Data Understanding (EDA – Exploratory Data Analysis)

- Check structure of data
- Identify missing values
- Visualize patterns and relationships

4 Data Cleaning & Preprocessing

- Handle missing values
- Remove duplicates
- Encode categorical variables
- Feature scaling

5 Feature Engineering

- Create new useful features
- Select important variables

6 Model Selection

- Choose appropriate ML algorithm
  - Regression
  - Classification
  - Clustering

7 Model Training

- Split data (Train/Test)
- Train model on training data

8 Model Evaluation

- Use metrics like:
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - RMSE

9 Model Deployment

- Deploy model using:
  - Flask / FastAPI
  - Streamlit
  - Cloud services

10 Monitoring & Maintenance

- Track performance
- Retrain when needed

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3. Machine Learning Problem Statement

Problem Title: Customer Churn Prediction

Business Problem:

Customer churn reduces company revenue. The company wants to identify customers who are likely to stop purchasing.

Objective:

Build a machine learning model that predicts whether a customer will churn based on their purchase behavior.

Input Variables (Features):

- Purchase frequency
- Total spending
- Last purchase date
- Average order value
- Customer tenure
- Discount usage

Output Variable (Target):

- Churn (Yes / No)

Type of Problem:

Binary Classification

Success Criteria:

- Achieve high Recall (to correctly identify churn customers)
- Improve customer retention strategy

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4. What are Tensors?

Definition:

A **Tensor** is a multi-dimensional array used in deep learning and scientific computing.

It is a generalization of:

- Scalar (0D Tensor) → Single number
- Vector (1D Tensor) → List of numbers
- Matrix (2D Tensor) → Table of numbers
- Higher Dimensions → 3D, 4D, etc.

In-Depth Explanation of Tensors with NumPy Examples

In Python, tensors can be created using **NumPy arrays**.

0️⃣ 0D Tensor (Scalar)

```
import numpy as np

scalar = np.array(10)
print(scalar)
print(scalar.ndim)    # 0
```

Output:

```
10
0
```

1️⃣ 1D Tensor (Vector)

```
vector = np.array([1, 2, 3, 4])
print(vector)
print(vector.ndim)    # 1
```

Output:

```
[1 2 3 4]
1
```

2️⃣ 2D Tensor (Matrix)

```
matrix = np.array([[1, 2],
                   [3, 4]])
print(matrix)
print(matrix.ndim)    # 2
```

Output:

```
[[1 2]
 [3 4]]
2
```

3️⃣ 3D Tensor

```
tensor3d = np.array([[[1, 2],
                      [3, 4]],

                     [[5, 6],
                      [7, 8]]])

print(tensor3d)
print(tensor3d.ndim)    # 3
```

Why Tensors Are Important?

In Deep Learning:

- Images → 3D tensors (Height × Width × Channels)
- Videos → 4D tensors
- Dataset batches → 2D or 3D tensors

Frameworks like:

- TensorFlow
- PyTorch

use tensors as the main data structure.